

No.	Solution	Remarks
2(a)	Body is moving in a circle/circular arc. The velocity is changing therefore there is an acceleration towards the centre of the circle/circular arc.	
2(b)(i)		
2(b)(ii)	At Y, for the ball to be just in contact with the loop, let the normal contact force be zero. $N + W = \frac{mv^2}{R}$ $mg = \frac{mv^2}{R}$ $v = \sqrt{gR}$	
2(b)(iii)	By Conservation of Energy, $mgH = mg(2R) + \frac{1}{2}m(\sqrt{gR})^2$ $H = 2.5R$	
2(c)	The rocket experiences equal and opposite gravitational	